IoT-Based Weed Detection & Removal System

Full Project Architecture (Phase 1 – Phase 6)

Phase 1: UI + Backend Setup

```
Frontend (HTML/CSS/JS)
        ↓ fetch()
Backend (Node.js API)
        ↓
Returns data (status/logs)
```

Phase 2: IoT Integration

```
ESP32 (WiFi Enabled)
        ↓ HTTP POST
Backend (/api/update)
        ↓
Updates system status
```

Phase 3: Database (Storage)

```
ESP32 → Backend
        ↓
logs.json / Database
        ↓
Frontend logs page
```

Phase 4: Automation (Control System)

```
Frontend (Settings Page)
        ↓
/api/control
        ↓
Backend
        ↓
ESP32 (Motor / Actuator)
```

Phase 5: Smart Detection (AI)

```
Camera Module
        ↓
Python + OpenCV / ML Model
        ↓
Detect weed
        ↓
Send result → Backend
```

Phase 6: Deployment & Finalization

```
Frontend → Vercel / Netlify
Backend → Render / Railway
Database → MongoDB / Firebase
System → Live Deployment
```

Complete System Flow

```
Camera/Sensors → ESP32 → Backend → Database → Frontend Dashboard
                              ↓
                       Control Commands
                              ↓
                           Actuators
```